



6. Programme Structure (Total Credits : 80 Credits)

1. Professional Core Courses
(C)
(10 Courses)

Course Code	Course Title	Hours/ Week			C
		L	T	P	
PCA25C01J	Object Oriented Programming using Java	3	0	2	4
PCA25C02T	Data structures and Algorithms	4	0	0	4
PCA25C03J	Database Technology	3	0	2	4
PCA25C04J	Data Analysis using Python Programming	3	0	2	4
PCA25C05T	Optimization Techniques	4	0	0	4
PCA25C06J	Advanced Operating System	3	0	2	4
PCA25C07J	Artificial Intelligence and Machine Learning	3	0	2	4
PCA25C08J	Data Communication Networks	3	0	2	4
PCA25C09T	Cloud and Quantum Computing	4	0	0	4
PCA25C10L	Capstone Project	0	0	8	4
Total Learning Credits					40

2. Discipline Elective
Courses (D)
(3 Courses)

Course Code	Course Title	Hours/ Week			C
		L	T	P	
PCA25D01J	Software Engineering and Testing	3	0	2	4
PCA25D02J	Advanced Web Technology				
PCA25D03J	Natural Language Processing				
PCA25D04J	Software Project Management	3	0	2	4
PCA25D05J	Mobile Applications Development				
PCA25D06J	Data Visualization Techniques				
PCA25D07J	Software Quality Assurance	3	0	2	4
PCA25D08J	Augmented Reality and Virtual Reality				
PCA25D09J	Computer Vision				
Total Learning Credits					12

3. Generic Elective Courses
(G)
(2 Courses)

Course Code	Course Title	Hours/ Week			C
		L	T	P	
PCA25G01T	Cyber Security	2	0	0	2
PCA25G02T	Soft Computing				
PCA25G03T	Distributed Computing				
PCA25G04T	Cyber Forensics	2	0	0	2
PCA25G05T	BlockChain Technology				
PCA25G06T	Big Data Analytics				
Total Learning Credits					4

4. Skill Enhancement Courses
(S)
(2 Courses)

Course Code	Course Title	Hours/ Week			C
		L	T	P	
PCA25S01J	Data Mining Techniques	2	0	3	4
PCA25S02J	Internet of Things (IoT)	2	0	3	4
Total Learning Credits					8

5. Project Work, Internship in
Industry/Higher Technical
Institutions (P)
(3 Courses)

Course Code	Course Title	Hours/ Week			C
		L	T	P	
PCA25P01L	Internship	0	0	0	2
PCA25P02L	Project Work	0	0	20	10
Total Learning Credits					12

6. Ability Enhancement
Courses (AE)
(2 Courses)

Course Code	Course Title	Hours/ Week			C
		L	T	P	
PCD25AE1T	Comprehensive Skills in Quantitative and Logical Reasoning	2	0	0	2
PCD25AE2T	Soft Skills and Verbal Mastery	2	0	0	2
Total Learning Credits					4



Course Code	Course Title	Hours/Week			
		L	T	P	C
PCA25P01L	Internship	0	0	0	2
PCA25P02L	Project Work	0	0	20	10
Total Learning Credits					12

Course Code	Course Title	Hours/Week			
		L	T	P	C
PCD25AE1T	Comprehensive Skills in Quantitative and Logical Reasoning	2	0	0	2
PCD25AE2T	Soft Skills and Verbal Mastery	2	0	0	2
Total Learning Credits					4



7. Implementation Plan						
Semester – I						
Code	Course Title	Hours/ Week			C	
		L	T	P		
PCA25C01J	Object Oriented Programming using Java	3	0	2	4	
PCA25C02T	Data structures and Algorithms	4	0	0	4	
PCA25C03J	Database Technology	3	0	2	4	
PCA25D01J	Software Engineering and Testing	3	0	2	4	
PCA25D02J	Advanced Web Technology	3	0	2		
PCA25D03J	Natural Language Processing	3	0	2		
PCA25G01T	Cyber Security	2	0	0	2	
PCA25G02T	Soft Computing	2	0	0		
PCA25G03T	Distributed Computing	2	0	0		
PCA25S01J	Data Mining Techniques	2	0	3	4	
PCD25AE1T	Comprehensive Skills in Quantitative and Logical Reasoning	2	0	0	2	
Total		28			24	

Semester - II						
Code	Course Title	Hours/ Week			C	
		L	T	P		
PCA25C04J	Data Analysis using Python Programming	3	0	2	4	
PCA25C05T	Optimization Techniques	4	0	0	4	
PCA25C06J	Advanced Operating System	3	0	2	4	
PCA25C07J	Artificial Intelligence and Machine Learning	3	0	2	4	
PCA25D04J	Software Project Management	3	0	2	4	
PCA25D05J	Mobile Applications Development	3	0	2		
PCA25D06J	Data Visualization Techniques	3	0	2		
PCA25S02J	Internet of Things (IoT)	2	0	3	4	
PCD25AE2T	Soft Skills and Verbal Mastery	2	0	0	2	
Total		30			26	

Semester – III						
Code	Course Title	Hours/ Week			C	
		L	T	P		
PCA25C08J	Data Communication Networks	3	0	2	4	
PCA25C09T	Cloud and Quantum Computing	4	0	0	4	
PCA25C10L	Capstone Project	0	0	8	4	
PCA25D07J	Software Quality Assurance	3	0	2	4	
PCA25D08J	Augmented Reality and Virtual Reality	3	0	2		
PCA25D09J	Computer Vision	3	0	2		
PCA25G04T	Cyber Forensics	2	0	0	2	
PCA25G05T	BlockChain Technology	2	0	0		
PCA25G06T	Big Data Analytics	2	0	0		
PCA25P01L	Internship	-	-	-	2	
Total		25			20	

Semester – IV						
Code	Course Title	Hours/ Week			C	
		L	T	P		
PCA25P02L	Project Work	0	0	20	10	
Total		20			10	